Engineering Economy- final Exam - Time: 90 min.

Student Name..... اسم الطالب

+++Use the DATA in the table above WHERE NEEDED. *** SHOW YOUR CALCULATIONS

Items	Alt.A	Alt.B	Alt.C	Alt.D	Alt.E	Alt. F
First. Cost	15,000	7,000	8,000	7,000	4500	2000
Annual Disbenefits A.D	-	-	-	2000	-	-
O.C. started from the 3rd year	1100	-	-	-	-	
O.C	-	-	-	500	-	-
Annual income / benefits	-	-	300	1000	240	450
Cost every 3 years	-	50	-	-	-	
Extra cost every 5 yrs	60	-	-	-	-	
Annual income for the first 4yrs.decreasing by G=50	-	100	200	-	350	
S.V	-	500	700	3,000	300	1000
Life(n) yrs	∞	7	8	5	6	6
i%	8%	12%	8%	8%	8%	MARR=20%

Q1-1) The Simple (I) that makes JD 1500 now equivalent to JD 2100 after 7 years is [ ]

1-2) How long will it take to make JD 1700 equivalent to JD 2500 if i=8% per year. [ ]

1-3) If in12% per year, compounded every 2months, then ieff. Semiannnally is: [ ]

Q2-1) The Future Worth for alternative FW (E) is: [ ]

2-2) The Equivalent (G) at the last 4 years for the alternative B is... [ ]

2-3) The Capitalized Cost for the Alternative CC (E) is ..... [ ]

Q3: Using the SOYD Depreciation for the alternative E:

3-1) The book value in the second year is: ..... [ ]

3-2) The book value in the 4th year is ..... [ ]

3-3) The depreciation cost in the 6th year is ..... [ ]

Q4: For the Alternatives A and C, Using AW criterion:-

a- AW (A) is ..... [ ]

b- AW(C) is ..... [ ]

c- The decision is ..... [ ]

Q5: For Alt. D

a- The modified B/C is ..... [ ]

b- You decision is ..... [ ]

Q6: For the alternative (F) find:

a- IRR i*(F) is ..... [ ]

b- Your decision is: ..... [ ]

Good Luck

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ENGINEERING ECONOMY
Final Exam

Time:- 90 Minutes

- Use the Data in the Table Below for all Given Questions

Items	Alt.A	Alt.B	Alt.C	Alt. D	Alt.E	Alt.
First (Initial) Cost	25000	32000	15000	17000	45000	40
Extra cost at the end of 4th yr.	2000	-	-		-	
Operating Cost	1200	1000	1500	1600	-	
Annual income / benefits	-	-	-	3500	-	7
Annual Users' Disbenefits	-		-	1800	-	
Salvage Value	5000	6500	3,000	2500	3000	1
Life(n) yrs	6	∞	6	5	6	
i%	9%	9%	9%	9%	9%	MA

Questions	Answ
1. For the alternative (A) find Present worth value?	1.
2. To compare Alternatives (A) and (B). Using AW criterion: a- AW (A) is b- AW (B) c- The decision is	a- b- c-
3. Find Capitalized Cost (CC) for alternative (C):	3.
4. For the alternative (D) Find : a-The conventional B/C is b-The modified B/C is c- Your decision is	a- b- C-
5. Using the (SOYD) depreciation method for alternative E : a-The book value in the second year is b-The book value in the 4th year is c-The depreciation cost in the 5th year is	a- b- c-
6. For the Alternative (F) find : a-IRR i* (F) is b-Your decision is	a- b-
7. Given *i. =18%* per year, compounded monthly. FIND *i.eff* per year	7.
8. Given *i.=24%* per year, compounded continuously. FIND *i.eff* per quarter	8.
9. If *i. = 6%* per month. FIND nominal interest rate per quarter	9

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